

Swarup K Subudhi

8704 Rhode Island Ave, College Park, MD

Ph: (240)-906-94345 | Email: swarupks@umd.edu

Website: swarupsubudhi.github.io/website

Experience

Graduate Research Assistant III, 2022-present

- **Development of Three-Dimensional High-Energy X-Ray Photoelectric Polarimeter Assembly Using Additive Manufacturing Technology**
Sponsor: NASA-Goddard Space Flight Center, Jan 2024 – present
- **Thermal Cycling Reliability of Printed Electronics**
Sponsor: Laboratory of Physical Sciences, June 2023 – Dec 2023
- **Reliability Validation and Sustainment of Direct-Write Printed RF Devices**
Sponsor: NextFlex, Jan 2022 – May 2023

Senior Manager, Manufacturing (Transmissions), 2019-20

Tata Motors Ltd. CVBU, Jamshedpur

- Improved Operational Line Efficiency of Gearbox Assembly Line by improving takt time & process quality parameters through process planning optimization
- Implemented low-cost ANDON analysis to capture assembly line losses through integration of IoT based tools with existing assembly line equipment & implemented loss-reduction plan

Assistant Manager, NPI (Transmissions), 2016-19

Tata Motors Ltd. CVBU, Jamshedpur

- Managed delivery of auxiliary gearboxes for 6X6 and 8X8 defense heavy trucks
- Launched ten new gearboxes (as per market requirements) and integrated into mass production through implementation of change-management processes

Education

Ph.D., Mechanical Engineering (exp. May 2026)

University of Maryland, College Park, MD, USA

Dissertation: Magnetic Films & Drops: And their Interactions with Other Viscous Mediums

Courses: Mechanical Fundamentals of Electronic Components, Composite Materials, Interfacial Fluid Mechanics, Modelling Material Behavior, Probabilistic PoF & Accelerated Testing, etc.

M.S., Mechanical Engineering, 2021

University of Maryland, College Park, MD, USA

Graduate Teaching Assistant I (Jan 2021 – Dec 2021)
Courses: Fluid Mechanics ENME331, Mechanics I ENES102

Bachelor of Technology, Mechanical Engineering, 2016

National Institute of Technology, Rourkela, Odisha, India

Major: Mechanical Engineering, Minor: Mathematics

Courses: Kinematics & Dynamics of Machinery, Machine Design I & II, Tool Design, Digital Logic Design, Materials Technology

Skills

Machine/ Product Design ::

CAD: SolidWorks, Pro-E; **CAE/FEA:** ANSYS Mechanical, Workbench, Icepak, Abaqus CAE

CFD: Fluent, COMSOL; **CAM:** Autodesk 360, Solidworks CAM, nScriptPro, Slicer (CraftPro)

Metrology: GD&T, SEM, Laser/ Optical Microscopy, Surface Profiling

Fabrication: Laser Tooling, 3D Printing, Subtractive Machining, Mold Dev

Control: Arduino, PID, MATLAB GUI, Sensors; **Integration:** Prototyping, Circuit Design, Testing
Matrix Development

Professional Tools: DFA/DFM, BOM, PLM, Six-Sigma, DoE, PSR, PFMEA, NPI Process

Reliability Testing & Analysis ::

Material Characterization: CTE, DMA (Stress Tests), DSC

Rheology Characterization: Density, Viscosity, Surface Tension, zeta-Potential, Size Distribution

Accelerated Testing: Thermal Cycling, Temp-Humidity Bias, Vibration/ Deformational Loading

Statistical Analysis: Acceleration Factors, Failure Analysis, DoE, FEA, ANOVA, Minitab

Computation & Analysis ::

Software: Matlab & SimuLink (Optimization & Image Processing), Mathematica, JupyterLab

Dev Languages: Python (Basic + CV2), C, C++, HTML5, CSS3, JavaScript

Other ::

Professional Tools: Microsoft Office, PowerBI | **Design Tools:** Photoshop, Illustrator

Publications

- Subudhi, S.K., Zhao, B., Wang, X., Ting, J., Takeuchi, I., Dasgupta, A., and Das, S. (2024) **Flexible and twistable free-standing PDMS-magnetic-nanoparticle-based soft magnetic films with robust magnetic properties.** *Flexible and Printed Electronics*, 9(1), 015013.

- Subudhi, S.K., Chandel, G.R., Sivasankar, V., and Das, S. (2024) **Magnetic Nanoparticle Aggregation and Complete De-Encapsulation of Such Aggregates from a Liquid Drop Interior.** *ACS Appl. Mater. Interfaces* 2024, 16, 46, 64253–64263.
- Subudhi, S.K. and Das, S. (2023) **Reliability of Lab-on-a-Chip Technologies for Wearable Electronics: A Perspective.** *Frontiers in Sensors*, 4, 1283402.
- Zhao, B., Sivasankar, V., Subudhi, S.K., Sinha, S., Nikfarjam, S., Woehl, T., and Das, S. (2023) **Three-Dimensional Deposits from Drying Particle-Laden Drops.** *Langmuir* 39(28), 9773-9784.
- Zhao, B., Sivasankar, V.S., Subudhi, S.K., Dasgupta, S., and Das, S. (2023) **Printed carbon nanotube-based humidity sensors deployable on surfaces of widely varying curvatures.** *ACS Applied Nano Materials* 6(2), 1459- 1474.
- Zhao, B., Sivasankar, V.S., Subudhi, S.K., Sinha, S., Dasgupta, A., and Das, S. (2022) **Applications, fluid mechanics, and colloidal science of carbon-nanotube-based 3D printable inks.** *Nanoscale* 14(40), 14858-14894
- Dhamsania et. al. (2022) **Physically soft magnetic films and devices: fabrication, properties, printability, and applications.** *Journal of Materials Chemistry C* 10(17), 6563-6589.

Past Professional Projects

- Case Study on Value Engineering of Hydraulic Tipping Kit of Heavy Commercial Tippers
 - Optimization & Cost sustainability of hydraulic tipping kits by alternate sourcing of gear pumps
- Value Engineering Case Study on G1150 (9-speed gearbox) synchronizing parts
 - Application of new material/technology in sustaining quality & performance of synchronizing cones & improving quality performance of gearboxes and cost sustainability of transmissions
- Six-Sigma Project: Warranty Cost Reduction of Gearboxes due to Bearing Failures
 - Application of six-sigma tools/concepts and DoEs to reduce the cost of warranty due to failure of all bearings in a commercial vehicle transmission, applicable for 3-24MIS warranty parts
- Operational Line Efficiency Improvement of Gearbox Assembly Line
 - Improvement of Operational Line Efficiency of Gearbox Assembly Line through optimization of takt time & process quality parameters depending on occurrence of defects and ergonomics

Past Academic Projects

- Synthesis and Analysis of a Reconfigurable Manipulator Assembly based on five-bar linkage
- Two-zone mathematical modelling of performance and combustion formation in direct-injection diesel engine operating on carbon black-water-diesel slurry
- Statistical modelling and optimization of MRR (material removal rate) and taper angle in reverse EDM (electro-discharge machining) using response-surface methodology

- Chassis and bodywork design of an electric karting vehicle. Structural deformation and stress analysis at given payload; and working model validation

Positions of Responsibilities

- ENME Representative to Graduate Student Government at UMD; Chair of Academic Affairs Committee, *University of Maryland, Jul 2022 – May 2025*
- Executive Body of Mechanical/Reliability Engineering Graduate Association, *University of Maryland, Jul 2024 – present*
- Leader of 3-5 cross-functional team projects, *Tata Motors Ltd. circa 2019*
- *To be updated...*